Machine learning Techniques

UNIT - 3 [One Shot]

Quick revision:

1. Decision Tree learning algorithm.
2. Inductive bias and Information gain.
3. ID-3 Algorithm
4. KNN algorithm and case-based learning.

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ⓐ Decision tree learning algorithm:

- It is a popular machine learning algorithm used for classification and regression tasks.
- Process of building a decision tree:

1. Choose the best feature:
   - Use a metric (information gain or gini index) to select the feature that best splits the data
2. Split the dataset:
   - Divide the dataset into subsets based on the chosen feature
3. Create decision node or leaf node:
   - If all samples belongs to one class, create a leaf node with class lable.
   - If all samples doesn't belongs to one class, go to step 2.
4. Repeat: Continue splitting recursively until stopping criteria are met such as maximum tree depth, minimum number of samples in a node or no further information gain.

Pruning in Decision tree:

It is used to receive the size of the tree by removing nodes that provide little powere in predicting

target variables, thereby improving model performance on unseen data.

Advantages:

- Easy to understand and interpret
- Requires little data preprocessing.
- Can handle both numerical and categorical data

Disadvantages:

- Prone to overfitting
- Can be unstable with small changes in data.
- Greedy nature can be lead to suboptimal splits.

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Inductive bias and information gain:

* Inductive bias: It refers to the set of assumptions that a learning algorithm uses to generalize from the training data to unseen instances.

- It helps to make predictions on new data by guiding the learning process in the face of limited information.
- Inductive bias is crucial because it helps the algorithm to generalize better by avoiding overfitting to the training data, allowing it to perform well on unseen data

* Information gain:

- It is a key metric used th to decide which feature to split on at each step in the construction of a decision tree.
- It measures the reduction in impurity achieved by partitioning the dataset according to a given attribute.

Calculation of Information gain:

1. **Entropy:** It is a measure of impurity or randomness in a dataset.

$$E(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

- S is the dataset
- c is no. of classes
- p_i is portion of instances.

2. **Conditional Entropy:** After splitting dataset based on an attribute, the conditional entropy is the weighted sum of the entropies of the resulting subsets.

$$E(S,A) = \sum_{j=1}^{n} \frac{|S_j|}{|S|} \text{Entropy}(S_j)$$

3. **Information gain:** It is the difference between the entropy of the original dataset and the conditional entropy after splitting on attribute A;

$$\text{Information gain}(S,A) = \text{Entropy}(S) - E(S,A)$$

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ID-3 algorithm:

- Iterative Dichotomiser 3 is a popular decision tree algorithm used in ML.
- It aims to build a decision tree by iteratively selecting the best attribute to split the data based on information gain.
- Each node represents a test on an attribute, and each branch represents a possible outcomes of the test.
- The leaf nodes of tree represent the final classifications.

Process:

1. Calculate Entropy of the Dataset:

- Determine the proportion of each class in the dataset.

2. Compute entropy for each attribute:

- Divide the dataset into subsets and calculate the entropy for each subset.

3. Calculate Information gain for each attribute:

- Compute the weighted average entropy for the subsets created by each attribute, and calculate information gain.

4. Select best attribute:

- Choose the attribute with the highest information gain as the decision node.

5. Create decision node:

- Use the selected attribute to split the dataset into subsets.
- Create branches for each possible value of the chosen attribute.

6. Repeat for each subset:

- Apply steps 1-5 until following conditions met:

  i) All instances in the subset belong to the same class.

  ii) There are no more attributes to split on

  iii) The dataset is empty

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#KNN Algorithm:

- K-Nearest Neighbors (KNN) algorithm is a simple, yet powerful supervised machine learning algorithm used for both classification and regression.
- It is based on the idea that similar data points are likely to be close to each other in feature space.

Steps in KNN algorithm:

1. Choose the number of neighbors (K):

- Decide the no of neighbors to consider (K).
- Common choices are small odd numbers (like 3 or 5) to avoid ties in classification.

2. Calculate the distance:
- Calculate Euclidean distance b/w given test data points.

3. Identify the K nearest neighbors:
- Sort calculated distance and select K- closest training data points.

4. Make a prediction:
- Find majority vote among the K- nearest neighbors, and assign the majority value as prediction for the the test data points.

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* Example:

X1	X2	Y
7	7	False
7	4	False
3	4	True
1	4	True
3	7	?

Step 1: let. $k = 3$ (always choose $k = 3$ or 5).

Step 2: Find Euclidean distance for each known Y, w.r.t. unknown X1 & X2.

- $ED(7,7),(3,7) = \sqrt{(7-3)^2 + (7-7)^2} = \sqrt{16} = 4$
- $ED(7,4),(3,7) = \sqrt{(7-3)^2 + (4-7)^2} = \sqrt{25} = 5$
- $ED(3,4),(3,7) = \sqrt{(3-3)^2 + (4-7)^2} = \sqrt{9} = 3$
- $ED(1,4),(3,7) = \sqrt{(1-3)^2 + (4-7)^2} = \sqrt{13} = 3.6$

Step 3: Majority check after arranging dataset in increasing order of ED, and choosing output(Y) of the 3 closest neighbors ($K=3$).

no.(True) = 2, no.(False) = 1, Hence, Y(3,7) = True.

Case-based Learning:

⇒ Used for classification and regression problem, process of solving new problems based on the solution of previous problems.

⇒ Case-based learning is an advanced instance based learning method which is used to solve more complex problem.

⇒ It does not use Euclidian Distance matrix.

⇒ When a new case arrives to classify then identical case is chick in memory if any similar cases is found then its solution is also retreived.

Steps in CBL:

- Retreive: Grather data from memory and chick previous similar solution to the problem.
- Reuse: Suggest the solution based on experience.
- Revise: Modifying or updating knowledge based on new information or feedback.
- Retain: Store this updated method in memory for future problems.

Applications:

1. Customer service help desk for diagnosis of problems.
2. Engg. and law for technical design and legal roles.
3. Medical science for patients case history and treatements.

Thanks for watching

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"All the best for your exams"